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2027 CYCLE

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CUET UG

Physics (322)

Wave Optics

Official Syllabus Unit 6 · Class 12 NCERT · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA Physics-322, Unit 6)	Key Exam Focus & Skills
Wavefront & Huygens' Principle	Wave front and Huygen's principle, reflection and refraction of plane waves at a plane surface using wave fronts.	Spherical, cylindrical, and plane wavefronts; secondary wavelets and forward envelope; behavior of wavefronts through lenses, prisms, and mirrors.
Proof of Reflection & Refraction	Proof of laws of reflection and refraction using Huygen's principle.	RHS triangle congruence proof for $i = r$; Snell's law derivation $\sin\frac{i}{\sin}r = \frac{v_1}{v_2} = \mu$; frequency invariance $\nu = \text{const}$ and wavelength compression $\lambda' = \frac{\lambda}{\mu}$.
Coherent Sources & Interference	Coherent sources and interference of light.	Conditions for sustained interference; superposition principle $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\varphi$; intensity extrema $\frac{I_{\max}}{I_{\min}} = \frac{(a_1+a_2)^2}{(a_1-a_2)^2}$.
Young's Double Slit Experiment	Young's double slit experiment and expression for fringe width.	Path difference $\Delta x = y\frac{d}{D}$; bright and dark fringe positions; fringe width $\beta = \lambda\frac{D}{d}$; angular fringe width $\theta = \frac{\lambda}{d}$; liquid medium $\beta' = \frac{\beta}{\mu}$; thin film optical path shift $\Delta y = \left(\frac{D}{d}\right)(\mu - 1)t$.
Single Slit Diffraction	Diffraction due to a single slit, width of central maxima.	Fraunhofer single slit minima $a\sin\theta = n\lambda$; secondary maxima $a\sin\theta = \left(n + \frac{1}{2}\right)\lambda$; central maximum width $\beta_0 = 2\lambda\frac{D}{a} = 2\beta$; comparison with interference.
Polarization & Brewster's Law	Polarization of light (CUET UG core application).	Transverse wave proof; unpolarized vs polarized light; Malus's law $I = I_0\cos^2\theta$; Brewster's law $\tan i_p = \mu$ with $i_p + r_p = 90°$; Rayleigh resolution criterion $\theta = 1.22\frac{\lambda}{D}$.

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EXAM PATTERN

CUET UG Exam Architecture: Unit 6 (Optics) carries the highest weightage in CUET UG Physics (approx 12% to 15%, 7 to 8 MCQs). Wave Optics contributes 3 to 4 direct conceptual and numerical questions, particularly focused on YDSE fringe width scaling in liquids, central diffraction maximum width, ratio of $\frac{I_{max}}{I_{min}}$, and Brewster's angle relations.

Wavefronts & Huygens' Principle

Concept of Wavefront

In wave theory, light travels through space in the form of continuous disturbances. > **Definition:** A **wavefront** is defined as the continuous locus of all points in a medium that vibrate in the exact same phase at any given instant.

- **Rays of Light:** A ray is an arrow drawn perpendicular to the wavefront pointing in the direction of wave propagation:

$$(k) \perp \text{ Wavefront}$$

Wavefront Type	Source Geometry	Shape of Wavefront	Intensity Dependence (I)
Spherical	Point source at finite distance	Concentric spheres	$I \propto \frac{1}{r^2}$ (Amplitude $A \propto \frac{1}{r}$)
Cylindrical	Linear slit source at finite distance	Coaxial cylinders	$I \propto \frac{1}{r}$ (Amplitude $A \propto \frac{1}{\sqrt{r}}$)
Plane	Source at infinite distance (e.g., Sun)	Parallel planes	$I =$ constant (Amplitude $A =$ constant)

Huygens' Principle of Secondary Wavelets

Christian Huygens in 1678 proposed a geometrical construction to determine the position and shape of a wavefront at any subsequent time:

1. **Every point on a primary wavefront** acts as a fresh source of secondary spherical disturbances, termed **secondary wavelets**.
2. These secondary wavelets spread out in all directions in the medium with the characteristic wave speed v.
3. **Forward Tangential Envelope:** A surface tangential to all these secondary spherical wavelets in the forward direction at time $t + \Delta t$ represents the new wavefront.
4. **Back-Wave Absence:** Huygens assumed secondary wavelets have maximum intensity in the forward direction and zero in the backward direction (formally proven by Kirchhoff using inclination factor $\frac{1+\cos\theta}{2}$).

Behavior of Plane Wavefronts Through Optical Elements

1. **Through a Thin Prism:** A plane wavefront incident on a prism emerges as a tilted plane wavefront because light travels faster through the thin apex of the prism than through the thick base.
2. **Through a Convex Lens:** A plane wavefront incident on a convex lens emerges as a **converging spherical wavefront** centered at the principal focus F, because the central rays traverse maximum glass thickness and suffer maximum time delay.
3. **Through a Concave Mirror:** A plane wavefront reflecting from a concave mirror emerges as a **converging spherical wavefront** converging toward the focal point F.

WORKED EXAMPLE 1: WAVEFRONT SHAPE AND INTENSITY SCALING

Problem: (a) State the shape of the wavefront emerging from a point source placed at the focus of a convex lens. (b) If the distance from a point source is doubled, how do the wave amplitude and light intensity change?

Solution: (a) A point source at the focus of a convex lens emits a diverging spherical wavefront. After refraction through the lens, all rays emerge parallel to the principal axis. Hence, the emerging wavefront is a **Plane Wavefront**. (b) For a spherical wave emitted by a point source:

$$I \propto \frac{1}{r^2} \Longrightarrow \frac{I_2}{I_1} = \left(\frac{r_1}{r_2}\right)^2 = \left(\frac{r_1}{2r_1}\right)^2 = \frac{1}{4}$$

The intensity reduces to $\frac{1}{4}$**th** of its initial value. Since intensity is proportional to the square of amplitude ($I \propto A^2$):

$$A \propto \frac{1}{r} \Longrightarrow \frac{A_2}{A_1} = \frac{r_1}{r_2} = \frac{1}{2}$$

The amplitude is **halved**.

Huygens' Proofs of Reflection & Refraction Laws

Proof of the Law of Reflection by Huygens' Principle

Consider a plane wavefront AB incident at an angle i on a plane reflecting mirror XY.

- At time $t = 0$, end A touches the mirror, while end B is at distance BC from point C on the mirror.
- In time t, the ray from B reaches C with speed v:

$$BC = vt$$

- In the same time t, secondary wavelets from A expand into a hemisphere of radius $AE = vt$ in the reflecting medium.
- Draw tangent CE from C to this wavelet. Then CE is the **reflected plane wavefront**, and ray AE is the reflected ray inclined at angle of reflection r.

GEOMETRIC CONGRUENCE PROOF OF REFLECTION

In right-angled triangles $\triangle ABC$ and $\triangle AEC$:

1. $\angle ABC = \angle AEC = 90°$ (Rays are perpendicular to wavefronts).
2. $AC = AC$ (Common hypotenuse).
3. $BC = AE = vt$ (Distances travelled in identical time t).

$$\therefore \triangle ABC \cong \triangle AEC \quad \text{[RHS Congruence Criterion]}$$

Consequently, corresponding angles are equal:

$$\angle BAC = \angle ECA \implies i = r$$

Since the incident wavefront, reflecting mirror, and reflected wavefront are perpendicular to the plane of paper, the incident ray, normal, and reflected ray all lie in the same plane.

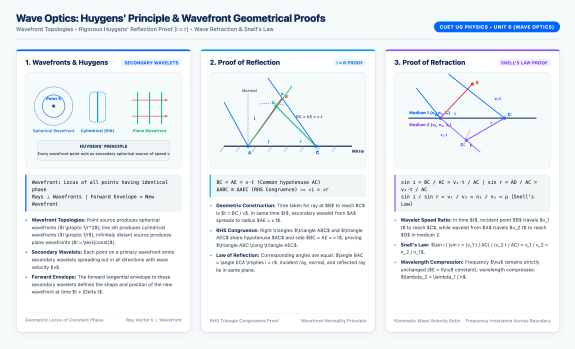


Figure 1: Huygens' Principle in Action: Wavefront topologies, RHS triangle congruence reflection proof, and Snell's refraction law derivation.

Proof of Snell's Law of Refraction by Huygens' Principle

Consider a plane wavefront AB incident at angle i on a planar interface XY separating medium 1 (speed v_1, refractive index n_1) from medium 2 (speed v_2, refractive index n_2).

- In time t, point B travels distance $BC = v_1 t$ in medium 1.
- In the same time t, secondary wavelets from A spread into medium 2 to a radius $AD = v_2 t$.
- Tangent CD represents the **refracted plane wavefront**, making angle of refraction r with interface XY.

SNELL'S LAW DERIVATION

From right triangle $\triangle ABC$:

$$\sin i = \frac{BC}{AC} = \frac{v_1 t}{AC}$$

From right triangle $\triangle ADC$:

$$\sin r = \frac{AD}{AC} = \frac{v_2 t}{AC}$$

Dividing the two equations:

$$\frac{\sin i}{\sin r} = \frac{\frac{v_1 t}{AC}}{\frac{v_2 t}{AC}} = \frac{v_1}{v_2}$$

By definition of absolute refractive index ($n_1 = \frac{c}{v_1}, n_2 = \frac{c}{v_2}$):

$$\frac{\sin i}{\sin r} = \frac{\frac{c}{n_1}}{\frac{c}{n_2}} = \frac{n_2}{n_1} = {}_1n_2 = \mu \quad \text{[Snell's Law]}$$

Frequency Invariance Across Boundary: The frequency ν of light is determined strictly by the oscillating source and remains **strictly constant** during refraction ($E = h\nu$). Consequently, wave speed and wavelength change in exact proportion:

$$\frac{v_1}{v_2} = \frac{\nu\lambda_1}{\nu\lambda_2} = \frac{\lambda_1}{\lambda_2} = \frac{n_2}{n_1} \Longrightarrow \lambda_2 = \frac{\lambda_1}{\mu}$$

WORKED EXAMPLE 2: REFRACTION PARAMETERS ACROSS GLASS INTERFACE

Problem: Monochromatic yellow light of wavelength $\lambda = 600$ nm in vacuum enters a glass slab of refractive index $\mu = 1.50$. Calculate: (a) the speed of light in glass, (b) the frequency of light in glass, and (c) the wavelength of light in glass.

Solution: (a) Speed of light in glass:

$$v = \frac{c}{\mu} = \frac{3.0 \times 10^8}{1.50} = 2.0 \times 10^8 \text{ m/s}$$

(b) Frequency in vacuum and glass (frequency is invariant!):

$$\nu = \frac{c}{\lambda} = \frac{3.0 \times 10^8}{600 \times 10^{-9}} = \frac{3.0 \times 10^8}{6.0 \times 10^{-7}} = 5.0 \times 10^{14} \text{ Hz}$$

Frequency in glass remains 5.0×10^{14} Hz. (c) Wavelength in glass:

$$\lambda' = \frac{\lambda}{\mu} = \frac{600 \text{ nm}}{1.50} = 400 \text{ nm}$$

(Verification: $v = \nu\lambda' = (5.0 \times 10^{14}) \times (400 \times 10^{-9}) = 2.0 \times 10^8$ m/s**).**

Coherent Sources & Interference of Light

Principle of Superposition

When two or more light waves propagate simultaneously through a medium, the resultant displacement (y) at any point is the vector sum of individual displacements:

$$(y) = (y)_1 + (y)_2 + \cdots + (y)_n$$

Coherent Sources

Two light sources are said to be **coherent** if they emit light waves of the same frequency (or wavelength) with a zero or constant, time-invariant phase difference ($\varphi = \text{constant}$).

- **Why Two Independent Sources Cannot be Coherent:** Light emission occurs via spontaneous atomic transitions lasting $\approx 10^{-8}$ s. Independent sources undergo billions of random, uncorrelated phase jumps every second, washing out interference fringes into uniform illumination.
- **Methods of Producing Coherent Sources:**

1. **Division of Wavefront:** A single primary wavefront is divided into two parts using double slits (Young's method), biprisms (Fresnel's biprism), or Lloyd's single mirror.
2. **Division of Amplitude:** Amplitude of an incident wave is split by partial reflection and partial refraction (e.g., thin soap films, Newton's rings).

Mathematical Theory of Interference

Consider two coherent waves of amplitudes a_1, a_2 and phase difference φ:

$$y_1 = a_1 \sin(\omega t), \quad y_2 = a_2 \sin(\omega t + \varphi)$$

Resultant displacement: $y = A\sin(\omega t + \theta)$ where resultant amplitude squared is:

$$A^2 = a_1^2 + a_2^2 + 2a_1 a_2 \cos\varphi$$

Since light intensity is proportional to amplitude squared ($I \propto A^2$):

RESULTANT INTENSITY FORMULA

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\varphi$$

If two identical slits are used ($I_1 = I_2 = I_0$):

$$I = 2I_0 + 2I_0\cos\varphi = 2I_0(1+\cos\varphi) = 4I_0\cos^2\left(\frac{\varphi}{2}\right)$$

Constructive Interference (Bright Fringes):

- Occurs when waves arrive in phase: $\cos\varphi = +1 \implies \varphi = 2n\pi$ $(n = 0, 1, 2, ...)$.
- Path difference: $\Delta x = n\lambda$.
- Maximum intensity: $I_{\max} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2 = (a_1 + a_2)^2$ (For identical slits: $I_{\max} = 4I_0$).

Destructive Interference (Dark Fringes):

- Occurs when waves arrive in anti-phase: $\cos\varphi = -1 \implies \varphi = (2n-1)\pi$ $(n = 1, 2, ...)$.
- Path difference: $\Delta x = \left(n - \frac{1}{2}\right)\lambda$.
- Minimum intensity: $I_{\min} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2 = (a_1 - a_2)^2$ (For identical slits: $I_{\min} = 0$).

Ratio of Maximum to Minimum Intensity

Slit width w is directly proportional to intensity and amplitude squared ($w \propto I \propto a^2$):

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{a_1 + a_2}{a_1 - a_2}\right)^2 = \left(\frac{\frac{a_1}{a_2} + 1}{\frac{a_1}{a_2} - 1}\right)^2 = \left(\frac{\sqrt{\frac{I_1}{I_2}} + 1}{\sqrt{\frac{I_1}{I_2}} - 1}\right)^2 = \left(\frac{\sqrt{\frac{w_1}{w_2}} + 1}{\sqrt{\frac{w_1}{w_2}} - 1}\right)^2$$

WORKED EXAMPLE 3: SLIT WIDTH RATIO & FRINGE CONTRAST

Problem: In a two-slit interference experiment, the ratio of the widths of the two slits is $\frac{w_1}{w_2} = \frac{4}{1}$. Calculate the ratio of maximum to minimum intensity in the resulting interference pattern.

Solution: The intensity transmitted by a slit is directly proportional to its width:

$$\frac{I_1}{I_2} = \frac{w_1}{w_2} = \frac{4}{1} \implies \frac{a_1}{a_2} = \sqrt{\frac{I_1}{I_2}} = \sqrt{\frac{4}{1}} = \frac{2}{1}$$

Ratio of maximum to minimum intensity:

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{a_1 + a_2}{a_1 - a_2}\right)^2 = \left(\frac{2+1}{2-1}\right)^2 = \left(\frac{3}{1}\right)^2 = \frac{9}{1}$$

Thus, $I_{\max} : I_{\min} = 9 : 1$.

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Young's Double Slit Experiment (YDSE)

Experimental Setup & Path Difference

Thomas Young (1801) demonstrated the wave nature of light using two narrow parallel slits S_1 and S_2 separated by distance d, illuminated by a single monochromatic pinhole source S.

- Screen is placed parallel to the slit plane at a large distance D ($D \gg d$).
- Consider an arbitrary point P on the screen at distance y from the central axis O.
- Path difference between waves arriving at P from S_2 and S_1:

$$\Delta x = S_2P - S_1P$$

Using geometry: $(S_2P)^2 - (S_1P)^2 = \left[D^2 + \left(y + \frac{d}{2}\right)^2\right] - \left[D^2 + \left(y - \frac{d}{2}\right)^2\right] = 2yd.$

$$(S_2P - S_1P)(S_2P + S_1P) = 2yd$$

Since $D \gg d$ and $y \ll D$, we can approximate $S_2P + S_1P \approx 2D$:

$$\Delta x = S_2P - S_1P = \frac{2yd}{2D} = \frac{yd}{D}$$

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POSITIONS OF FRINGES & FRINGE WIDTH

1. Positions of Bright Fringes (Constructive):

$$\Delta x = \frac{yd}{D} = n\lambda \Longrightarrow y_n = \frac{n\lambda D}{d} \quad (n = 0, \pm1, \pm2, ...)$$

- Central bright fringe ($n = 0$): $y_0 = 0$.
- First bright fringe ($n = 1$): $y_1 = \frac{\lambda D}{d}$.

2. Positions of Dark Fringes (Destructive):

$$\Delta x = \frac{yd}{D} = \left(n - \frac{1}{2}\right)\lambda \Longrightarrow y_{n'} = \left(n - \frac{1}{2}\right)\frac{\lambda D}{d} \quad (n = 1, 2, 3, ...)$$

- First dark fringe ($n = 1$): $y_{1'} = \left(\frac{1}{2}\right)\frac{\lambda D}{d}$.

3. Fringe Width (β): Separation between two consecutive bright or two consecutive dark fringes:

$$\beta = y_{n+1} - y_n = (n+1)\frac{\lambda D}{d} - \frac{n\lambda D}{d} = \frac{\lambda D}{d}$$

$$\beta' = y_{(n+1)'} - y_{n'} = \frac{\lambda D}{d} = \beta$$

Fundamental Rule: In interference, bright and dark fringes have strictly **identical widths**!



Figure 2: Interference and Diffraction Architecture: YDSE geometry, Fraunhofer single-slit central maximum width (2β), and Brewster's polarization angle.

Crucial Parametric Scalings in YDSE

1. **Angular Fringe Width (θ):**

$$\theta = \frac{\beta}{D} = \frac{\lambda}{d} \quad \text{[radians]}$$

Key Exam Fact: The angular fringe width is **independent** of the screen distance D!

2. **YDSE Immersed in a Liquid:** When the apparatus is immersed in a liquid of refractive index μ, the wavelength reduces to $\lambda' = \frac{\lambda}{\mu}$. Slit separation d and screen distance D remain unchanged:

$$\beta' = \frac{\lambda' D}{d} = \frac{\lambda D}{\mu d} = \frac{\beta}{\mu}$$

The fringe width **decreases** by factor μ.

3. **Displacement of Fringes Due to a Thin Transparent Plate:** When a thin transparent mica/glass sheet of thickness t and refractive index μ is placed in front of one slit (S_1), the optical path increases by $(\mu - 1)t$.
   - The entire fringe pattern shifts towards the covered slit by:

$$\Delta y = \frac{D}{d}(\mu - 1)t = \frac{\beta}{\lambda}(\mu - 1)t$$

   - **Crucial CUET Note:** The **fringe width β remains completely unchanged**; only the central position shifts!

4. **Illumination with White Light:** The central fringe ($y = 0$) has zero path difference for all wavelengths, appearing **pure white**. Fringes on either side appear coloured, with **violet** forming closest to the center (λ_{violet} shortest) and **red** farthest.

WORKED EXAMPLE 4: YDSE FRINGE WIDTH AND LIQUID IMMERSION

Problem: In a Young's double slit experiment using light of wavelength $\lambda = 600$ nm, the slit separation is $d = 0.50$ mm and screen distance is $D = 1.0$ m. (a) Find the fringe width. (b) Find the distance of the 4th bright fringe from the central maximum. (c) What is the new fringe width if the entire apparatus is immersed in water ($\mu = \frac{4}{3}$)?

Solution: (a) Fringe width in air:

$$\beta = \frac{\lambda D}{d} = \frac{(600 \times 10^{-9}) \times 1.0}{0.50 \times 10^{-3}} = \frac{6.0 \times 10^{-7}}{5.0 \times 10^{-4}} = 1.20 \times 10^{-3} \text{ m} = 1.20 \text{ mm}$$

(b) Distance of the 4th bright fringe ($n = 4$):

$$y_4 = 4\beta = 4 \times 1.20 \text{ mm} = 4.80 \text{ mm}$$

(c) Fringe width in water ($\mu = \frac{4}{3}$):

$$\beta' = \frac{\beta}{\mu} = \frac{1.20 \text{ mm}}{\frac{4}{3}} = 1.20 \times \frac{3}{4} = 0.90 \text{ mm}$$

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Fraunhofer Single-Slit Diffraction

Concept of Diffraction

Diffraction is the phenomenon of bending of light waves around the edges of sharp obstacles or apertures into the region of geometrical shadow.

- **Condition for Observable Diffraction:** The size of the aperture/slit a must be comparable to the wavelength of the wave ($a \approx \lambda$ or $a \leq \lambda$).
- Since visible light has extremely tiny wavelengths ($\lambda \approx 500$ nm $= 5 \times 10^{-7}$ m), everyday obstacles do not produce noticeable diffraction, unlike sound waves ($\lambda \approx 1$ m) which bend easily around doors and walls.

Single-Slit Fraunhofer Diffraction

A monochromatic parallel beam of wavelength λ is incident normally on a single slit of width a. Light diffracted at angle θ is focused onto a screen at distance D by a converging lens.

- The slit is divided into continuous secondary wavelets. The path difference between rays from opposite edges of the slit is:

$$\Delta x = a \sin\theta$$

SINGLE-SLIT DIFFRACTION CONDITIONS

1. Condition for Minima (Dark Fringes): Dividing the slit into $2n$ equal halves where opposite wavelet pairs cancel:

$$a \sin\theta_n = n\lambda \Longrightarrow \sin\theta_n \approx \theta_n = \frac{n\lambda}{a} \quad (n = \pm1, \pm2, \pm3, ...)$$

- First minimum: $\theta_1 = \frac{\lambda}{a}$.

2. Condition for Secondary Maxima (Bright Fringes): Dividing the slit into $2n + 1$ equal strips where one strip remains uncancelled:

$$a \sin\theta_{n'} = \left(n + \frac{1}{2}\right)\lambda \Longrightarrow \theta_{n'} \approx \left(n + \frac{1}{2}\right)\frac{\lambda}{a} \quad (n = 1, 2, 3, ...)$$

3. Width of the Central Maximum: The central maximum spreads between the first minimum on either side ($-\frac{\lambda}{a}$ to $+\frac{\lambda}{a}$):

- **Angular Width:**

$$2\theta_0 = 2\left(\frac{\lambda}{a}\right) = \frac{2\lambda}{a} \quad \text{[radians]}$$

- **Linear Width on Screen (β_0):**

$$\beta_0 = 2y_1 = 2(\theta_1 D) = \frac{2\lambda D}{a} = 2\beta$$

Crucial Rule: The central diffraction maximum is **twice as wide** as any secondary maximum!

Comparison: Interference vs Diffraction

Feature	Interference (e.g., YDSE)	Diffraction (e.g., Single Slit)
Origin	Superposition of waves from two distinct coherent slits	Superposition of secondary wavelets originating from different parts of the same single slit
Fringe Width	All fringes (central and secondary) have equal width: $\beta = \lambda\frac{D}{d}$	Central maximum is **twice** as wide as secondary maxima: $\beta_0 = 2\lambda\frac{D}{a}$
Intensity Distribution	All bright fringes possess identical peak intensity ($I_{\max} = 4I_0$)	Central maximum has highest intensity; secondary maxima rapidly decay ($I_0 : \frac{I_0}{22} : \frac{I_0}{61}$)
Dark Fringes	Complete darkness ($I_{\min} = 0$) when slit amplitudes are equal	Dark fringes are rarely completely black due to residual wavelet contributions

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WORKED EXAMPLE 5: SINGLE SLIT CENTRAL MAXIMUM LINEAR WIDTH

Problem: A screen is placed 2.0 m away from a single slit of width 0.20 mm. When illuminated by light of wavelength 500 nm, determine: (a) the angular width, and (b) the linear width of the central diffraction maximum.

Solution: Given: $a = 0.20$ mm $= 2.0 \times 10^{-4}$ m, $\lambda = 500$ nm $= 5.0 \times 10^{-7}$ m, $D = 2.0$ m. (a) Angular width of central maximum:

$$2\theta_0 = \frac{2\lambda}{a} = \frac{2 \times 5.0 \times 10^{-7}}{2.0 \times 10^{-4}} = \frac{10^{-6}}{2.0 \times 10^{-4}} = 5.0 \times 10^{-3} \text{ rad}$$

(b) Linear width on the screen:

$$\beta_0 = \frac{2\lambda D}{a} = (2\theta_0) \times D = (5.0 \times 10^{-3}) \times 2.0 = 1.0 \times 10^{-2} \text{ m} = 10 \text{ mm} = 1.0 \text{ cm}$$

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Polarization & Brewster's Law

Transverse Nature & Polarization

Light waves are transverse electromagnetic waves where the electric field vector (E) vibrates in all possible directions perpendicular to the direction of propagation. Such light is termed **unpolarized light**.

- **Linearly (Plane) Polarized Light:** Light in which electric field vibrations are confined strictly to a single plane perpendicular to the direction of propagation.
- **Proof of Transverse Nature:** Longitudinal waves (like sound in air) cannot be polarized because their vibrations are along the direction of propagation. Hence, the occurrence of polarization is the **definitive proof** that light is a transverse wave!

Malus's Law

Malus's Law & Polarization by Reflection

When completely plane-polarized light of intensity I_0 is incident on an analyzer inclined at angle θ:

MALUS'S LAW & UNPOLARIZED TRANSMISSION

$$I = I_0 \cos^2 \theta$$

- $\theta = 0°$: $I = I_0$ (parallel axes) | $\theta = 90°$: $I = 0$ (crossed polaroids, complete extinction).
- Unpolarized light on a polaroid: $I_1 = \frac{1}{2} I_{\text{unpolarized}}$ (since $\overline{\cos^2 \theta} = \frac{1}{2}$).

Brewster's Law & Polarization by Reflection

At Brewster's polarizing angle of incidence i_p, the reflected ray is 100% linearly polarized with vibrations perpendicular to the plane of incidence, and the reflected and refracted rays are perpendicular ($i_p + r_p = 90°$).

BREWSTER'S LAW RELATION

$$\mu = \frac{\sin i_p}{\sin r_p} = \frac{\sin i_p}{\cos i_p} = \tan i_p \Longrightarrow i_p = \tan^{-1}(\mu)$$

- For glass ($\mu = 1.50$): $i_p \approx 56.3°$ | For water ($\mu = 1.33$): $i_p \approx 53.1°$.

WORKED EXAMPLE 6: MALUS'S LAW WITH THREE POLAROIDS

Problem: Unpolarized light of intensity I_0 passes through two crossed polaroids P_1 and P_2. A third polaroid P_3 is inserted between them with its pass axis at 45° to P_1. Find the transmitted intensity from P_2.

Solution:

1. Transmitted through P_1: $I_1 = \frac{1}{2} I_0$.
2. Transmitted through P_3 ($\theta_1 = 45°$): $I_3 = I_1 \cos^2(45°) = \left(\frac{1}{2} I_0\right)\left(\frac{1}{2}\right) = \frac{1}{4} I_0$.
3. Transmitted through P_2 ($\theta_2 = 90° - 45° = 45°$): $I_2 = I_3 \cos^2(45°) = \left(\frac{1}{4} I_0\right)\left(\frac{1}{2}\right) = \frac{1}{8} I_0$.

Transmitted intensity is $\frac{I_0}{8}$. (Without P_3, transmitted intensity is zero).

Master Mindmap & High-Yield Cheatsheet



Figure 3: Master Synthesis Mindmap: Structural framework of Unit 6 Wave Optics.

MASTER WAVE OPTICS REVISION CHEATSHEET

- **Wavefront Types:** Point source: Spherical ($I \propto \frac{1}{r^2}$) | Slit source: Cylindrical ($I \propto \frac{1}{r}$) | Distant: Plane ($I = \text{const}$).
- **Refraction Invariance:** Frequency ν is invariant ($E = h\nu = \text{const}$) | $v = \frac{c}{\mu}$ | Wavelength compresses: $\lambda' = \frac{\lambda}{\mu}$.
- **Interference Extrema:** $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\varphi$ | $\frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{I_1}+\sqrt{I_2}}{\sqrt{I_1}-\sqrt{I_2}}\right)^2 = \left(\frac{a_1+a_2}{a_1-a_2}\right)^2$.
- **YDSE Fringe Width:** $\beta = \frac{\lambda D}{d}$ (uniform for bright and dark) | Angular width $\theta = \frac{\lambda}{d}$ (independent of D).
- **Liquid Immersion:** $\beta' = \frac{\beta}{\mu}$ | Mica film shift: $\Delta y = \left(\frac{D}{d}\right)(\mu - 1)t$ (fringe width β unchanged!).
- **Diffraction Minima:** $a\sin\theta = n\lambda$ | Secondary Maxima: $a\sin\theta = \left(n + \frac{1}{2}\right)\lambda$.
- **Central Max Width:** Linear: $\beta_0 = \frac{2\lambda D}{a} = 2\beta$ (twice secondary width!) | Angular: $2\theta_0 = \frac{2\lambda}{a}$.
- **Malus's Law:** $I = I_0\cos^2\theta$ | Unpolarized through single sheet: $I = \frac{I_0}{2}$.
- **Brewster's Law:** $\tan i_p = \mu$ | Perpendicular rays: $i_p + r_p = 90°$ (reflected ray is 100% linearly polarized).

CUET UG Exam-Style Practice Questions (15 MCQs)

Q1. When a monochromatic light wave of wavelength 600 nm travelling in vacuum enters a medium of refractive index 1.50, its frequency and wavelength in the medium are:

(A) 5.0×10^{14} Hz, 600 nm
(B) 5.0×10^{14} Hz, 400 nm
(C) 7.5×10^{14} Hz, 400 nm
(D) 3.33×10^{14} Hz, 900 nm

Sol. Q1. Frequency is invariant: $\nu = \frac{c}{\lambda} = \frac{3.0\times10^8}{600\times10^{-9}} = 5.0 \times 10^{14}$ Hz. Wavelength in medium compresses: $\lambda' = \frac{\lambda}{\mu} = \frac{600}{1.50} = 400$ nm.

Answer: (B)

Q2. In Young's double slit experiment, if the separation between the slits is halved and the distance between slits and screen is doubled, the fringe width becomes:

(A) Halved
(B) Doubled
(C) Four times
(D) Unchanged

Sol. Q2. Fringe width $\beta = \frac{\lambda D}{d}$. When $d' = \frac{d}{2}$ and $D' = 2D$: $\beta' = \frac{\lambda(2D)}{\frac{d}{2}} = 4\left(\lambda\frac{D}{d}\right) = 4\beta$. Fringe width quadruples.

Answer: (C)

Q3. Two coherent light sources having intensity ratio $\frac{I_1}{I_2} = \frac{9}{1}$ produce an interference pattern. The ratio of $\frac{I_{\max}-I_{\min}}{I_{\max}+I_{\min}}$ is:

(A) $\frac{3}{5}$
(B) $\frac{3}{10}$
(C) $\frac{4}{5}$
(D) $\frac{1}{3}$

Sol. Q3. $\frac{a_1}{a_2} = \sqrt{\frac{I_1}{I_2}} = \sqrt{9} = 3$. $I_{\max} = (3+1)^2 = 16$, $I_{\min} = (3-1)^2 = 4$. $\frac{I_{\max}-I_{\min}}{I_{\max}+I_{\min}} = \frac{16-4}{16+4} = \frac{12}{20} = \frac{3}{5}$.

Answer: (A)

Q4. In a Young's double slit experiment, the fringe width is found to be 0.40 mm. If the entire apparatus is immersed in water ($\mu = \frac{4}{3}$) without altering other parameters, the new fringe width is:

(A) 0.53 mm
(B) 0.30 mm
(C) 0.40 mm
(D) 0.20 mm

Sol. Q4. In medium of refractive index μ, fringe width scales as $\beta' = \frac{\beta}{\mu} = \frac{0.40 \text{ mm}}{\frac{4}{3}} = 0.40 \times \frac{3}{4} = 0.30$ mm.

Answer: (B)

Q5. A thin transparent sheet of thickness t and refractive index μ is introduced in the path of one of the interfering beams in a YDSE. The central fringe shifts by:

(A) $\frac{D}{d}(\mu-1)t$
(B) $\frac{d}{D}(\mu-1)t$
(C) $\frac{D}{d}\mu t$
(D) $\frac{D}{2d}(\mu-1)t$

Sol. Q5. The optical path length increases by $(\mu-1)t$. To compensate, the path difference $y\frac{d}{D} = (\mu-1)t \Longrightarrow y = \left(\frac{D}{d}\right)(\mu-1)t$.

Answer: (A)

Q6. In Fraunhofer diffraction by a single slit of width a, the first minimum for light of wavelength λ occurs at an angular position θ given by:

(A) $\sin\theta = \frac{\lambda}{2a}$ (B) $\sin\theta = \frac{\lambda}{a}$
(C) $\sin\theta = \frac{3\lambda}{2a}$ (D) $\sin\theta = \frac{2\lambda}{a}$

Sol. Q6. Condition for diffraction minima at single slit of width a is $a\sin\theta = n\lambda$. For the first minimum ($n = 1$), $\sin\theta = \frac{\lambda}{a}$.

Answer: (B)

Q7. The linear width of the central maximum in a single-slit Fraunhofer diffraction pattern on a screen at distance D from a slit of width a is:

(A) $\frac{\lambda D}{a}$ (B) $\frac{2\lambda D}{a}$
(C) $\frac{\lambda D}{2a}$ (D) $\frac{4\lambda D}{a}$

Sol. Q7. The central maximum extends between the first minima on either side ($-\frac{\lambda}{a}$ to $+\frac{\lambda}{a}$). Angular width is $2\frac{\lambda}{a}$, so linear width is $\beta_0 = \frac{2\lambda D}{a}$.

Answer: (B)

Q8. Which of the following optical phenomena conclusively proves that light waves are transverse and cannot be longitudinal?

(A) Interference (B) Diffraction
(C) Polarization (D) Total Internal Reflection

Sol. Q8. Only transverse waves can exhibit asymmetry of vibrations about the direction of propagation (polarization). Longitudinal waves cannot be polarized.

Answer: (C)

Q9. Unpolarized light of intensity I_0 passes through a Polaroid sheet. The intensity of transmitted light is:

(A) I_0 (B) $\frac{I_0}{2}$
(C) $\frac{I_0}{4}$ (D) $\frac{I_0}{\sqrt{2}}$

Sol. Q9. A polaroid transmits electric field components parallel to its axis. Averaging over all random incident angles ($\overline{\cos^2\theta} = \frac{1}{2}$), transmitted intensity is $I = \frac{I_0}{2}$.

Answer: (B)

Q10. Light is incident on a glass plate of refractive index $\mu = \sqrt{3}$ at Brewster's polarizing angle i_p. The angle of refraction r_p inside the glass is:

(A) 60° (B) 30°
(C) 45° (D) 90°

Sol. Q10. $\tan i_p = \mu = \sqrt{3} \Longrightarrow i_p = 60°$. At Brewster's angle, reflected and refracted rays are perpendicular: $i_p + r_p = 90° \Longrightarrow r_p = 90° - 60° = 30°$.

Answer: (B)

Q11. In a YDSE, the distance between 5th bright fringe and 2nd bright fringe on the same side of the central maximum is 6.0 mm. The fringe width is:

(A) 1.2 mm (B) 2.0 mm
(C) 3.0 mm (D) 1.5 mm

Sol. Q11. $y_5 - y_2 = (5-2)\beta = 3\beta = 6.0$ mm $\Longrightarrow \beta = \frac{6.0}{3} = 2.0$ mm.

Answer: (B)

Q12. If the single slit width a in a Fraunhofer diffraction experiment is doubled, the linear width of the central maximum on the screen will be:

(A) Doubled (B) Halved
(C) Four times (D) Unchanged

Sol. Q12. Linear width of central maximum is $\beta_0 = \frac{2\lambda D}{a} \propto \frac{1}{a}$. Doubling the slit width ($a' = 2a$) halves the width of the central maximum.

Answer: (B)

Q13. Two Nicol prisms are arranged with their transmission axes crossed ($\theta = 90°$). When the analyzer is rotated by 30°, the fraction of incident unpolarized light I_0 transmitted is:

(A) $\frac{I_0}{8}$ (B) $\frac{I_0}{4}$
(C) $\frac{3I_0}{8}$ (D) $\frac{I_0}{16}$

Sol. Q13. After polarizer: $I_1 = \frac{I_0}{2}$. Angle between axes is now $\theta = 90° - 30° = 60°$. $I = I_1 \cos^2(60°) = \left(\frac{I_0}{2}\right)\left(\frac{1}{2}\right)^2 = \frac{I_0}{8}$.

Answer: (A)

Q14. Rayleigh's criterion states that two point objects are just resolved by a circular aperture of diameter D when the angular separation θ is at least:

(A) $\frac{1.22\lambda}{D}$ (B) $\frac{\lambda}{D}$
(C) $\frac{2.44\lambda}{D}$ (D) $\frac{0.61\lambda}{D}$

Sol. Q14. By Airy disk diffraction resolution criterion, the central maximum of one image must fall on the first minimum of the other: $\theta = \frac{1.22\lambda}{D}$.

Answer: (A)

Q15. When white light is used in Young's double slit experiment, the fringe at the center of the screen ($y = 0$) appears:

(A) Dark (B) Bright Red
(C) Pure White (D) Bright Violet

Sol. Q15. At the center $y = 0$, the path difference is zero for all wavelengths, so all spectral colours interfere constructively, producing a central pure white fringe.

Answer: (C)